

Final Term Project Report

Project Title	Machine Learning-Based Student Academic Outcome Classification
Course / Section	[Programming in Python / D]
Instructor	MD. TANZEEM RAHAT
Date of Submission	12/09/2026

Group members:

ID	Name
23-51897-2	Alim Al Razi Shatak
24-58756-2	MD SHAZZADUL ISLAM
21-45527-3	Jannatun Fardous
23-50585-1	MD. Nahian Islam

1. Introduction

This project focuses on building and evaluating machine learning models for predicting the academic status of students. The target variable contains three possible classes: Graduate, Dropout, and Enrolled.

The main purpose of the project is to understand the dataset, prepare the data properly, train different machine learning models, compare their performance, and analyze the prediction results. The project follows a systematic workflow starting from project setup and dataset inspection and ending with final evaluation, error analysis, documentation, and GitHub cleanup.

The main classification models used in this project are Baseline, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM).

2. Project Workflow

The project was completed through the following steps:

1. Project and GitHub setup
2. Dataset and data provenance
3. Data audit
4. Exploratory Data Analysis (EDA) and visualizations
5. Train-test split
6. Data preprocessing
7. Baseline model
8. KNN model
9. SVM model
10. Model comparison
11. Final evaluation
12. Error analysis
13. Results and figures
14. README and requirements
15. Final report
16. Final GitHub cleanup

This workflow helps keep the project organized and makes the machine learning process easier to understand and reproduce.

3. Dataset and Data Provenance

The dataset contains information related to students and their academic outcomes. The target variable represents the student's current academic status.

There are three target classes:

- Graduate
- Dropout
- Enrolled

The dataset contains 4,424 records and 37 columns. Among the columns, 36 are numeric features, while one column represents the target variable.

The dataset was inspected before model development to understand its structure, feature types, missing values, duplicate records, and class distribution.

4. Data Audit

A data audit was performed before training the models. This step was important because problems such as missing values, duplicate records, or incorrect data types could affect the machine learning results.

Dataset Audit Results

Check	Result
Total Rows	4,424
Total Columns	37
Missing Values	0
Duplicate Rows	0
Numeric Features	36
Target	Target
Number of Classes	3

The audit shows that the dataset is relatively clean. There are no missing values and no duplicate rows. Therefore, no missing-value imputation or duplicate removal was required before continuing with the analysis.

The presence of 36 numeric features also makes the dataset suitable for applying numerical machine learning classification algorithms.

5. Target Class Distribution

The target variable contains three different student outcomes. Their distribution is shown below.

Original Class	Records	Percentage
Graduate	2,209	49.93%
Dropout	1,421	32.12%
Enrolled	794	17.95%
Total	4,424	100%

The Graduate class has the highest number of records, with 2,209 students, representing approximately 49.93% of the dataset.

The Dropout class contains 1,421 records, representing 32.12% of the dataset.

The Enrolled class is the smallest class, containing 794 records or 17.95% of the total data.

This shows that the dataset is not perfectly balanced. The Enrolled class has considerably fewer records than the Graduate class. Therefore, class distribution should be considered when evaluating machine learning models.

6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to better understand the characteristics of the dataset.

The analysis focused on examining the target class distribution and understanding the numerical features. Visualizations can help identify differences between classes and provide a better understanding of the data before applying machine learning algorithms.

The class distribution indicates that Graduate students form the largest group, while Enrolled students form the smallest group.

EDA is useful because it provides an initial understanding of the dataset before the models are trained. It can also help identify unusual patterns, potential outliers, and relationships between features and the target variable.

7. Train-Test Split

After completing the initial data analysis, the dataset was divided into training and testing portions.

The training data was used to train the machine learning models, while the testing data was kept separate for evaluating how well the trained models perform on unseen data.

Keeping the test data separate is important because evaluating a model on the same data used for training can produce misleading results.

A suitable train-test strategy was used to maintain a reliable representation of the three target classes in both sets.

8. Data Preprocessing

Before training the machine learning models, the feature data was prepared for classification.

Preprocessing is particularly important for distance-based and margin-based algorithms. For example, KNN depends on distances between observations, while SVM can also be affected by differences in feature scales.

The preprocessing stage therefore prepares the numerical features so that the models can work with the data more effectively.

Importantly, preprocessing steps should be fitted using the training data and then applied to the testing data. This helps prevent information from the test set from influencing the training process.

9. Baseline Model

A baseline model was used as a starting point for comparison.

The baseline achieved:

- **Accuracy: 49.94%**

- **Macro F1: 22.21%**

The baseline result provided a reference point for evaluating the more advanced models.

The low Macro F1 score shows that the baseline did not perform equally well across the three student outcome classes.

10. K-Nearest Neighbors

The second model used in the project was K-Nearest Neighbors (KNN).

KNN classifies a new student by examining the closest observations in the training dataset. The model identifies the nearest examples based on their feature values and assigns the student to the class that is most appropriate among those neighboring observations.

KNN is simple and easy to understand, but its performance can be affected by feature scaling and the choice of the number of neighbors.

The KNN model achieved:

- **Accuracy: 66.78%**
- **Macro F1: 57.30%**

Compared with the baseline, KNN provided a significant improvement in both accuracy and Macro F1.

This indicates that the feature information contained useful patterns for distinguishing between the three student outcomes.

11. Support Vector Machine

The third model used was Support Vector Machine (SVM).

SVM attempts to find a decision boundary that separates different classes effectively. For a multi-class problem such as this project, SVM can distinguish between the three student outcomes.

SVM is useful for classification problems with multiple numerical features. However, its performance can depend on preprocessing and suitable model parameters.

The SVM model achieved:

- **Accuracy: 75.82%**
- **Macro F1: 67.85%**

SVM produced the highest scores among the models tested in this project.

The result shows that SVM was able to distinguish the three student outcome classes more effectively than both the baseline and KNN models.

12. Model Comparison

The performance of the three models is summarized below.

Model	Accuracy	Macro F1
Baseline	49.94%	22.21%
KNN	66.78%	57.30%
SVM	75.82%	67.85%

The results show a clear improvement from the baseline to KNN and then to SVM.

SVM achieved the highest accuracy of 75.82% and the highest Macro F1 score of 67.85%.

Therefore, SVM was selected as the best-performing model in this project based on the reported evaluation of metrics.

13. Final Evaluation

The final evaluation focused on comparing the models using both Accuracy and Macro F1.

Accuracy measures the overall percentage of correctly classified student records. Macro F1 calculates the F1 score for each class and then gives equal importance to all classes.

This is useful for this dataset because the three target classes are not equally distributed.

The final results were:

- Baseline: **49.94% Accuracy, 22.21% Macro F1**
- KNN: **66.78% Accuracy, 57.30% Macro F1**
- SVM: **75.82% Accuracy, 67.85% Macro F1**

SVM performed better than the other tested models on both metrics. Its higher Macro F1 also indicates better overall balance across the three classes compared with the other models.

14. Error Analysis

The classification problem contains three possible outcomes: Graduate, Dropout, and Enrolled.

Because the classes have different numbers of records, some classes may be more difficult for a model to identify correctly.

The difference between Accuracy and Macro F1 is useful for understanding this issue. For example, the baseline achieved an accuracy close to 50%, but its Macro F1 was only 22.21%. This suggests that its performance was not balanced across the three classes.

KNN improved the results considerably, while SVM produced the strongest overall performance.

The remaining classification errors show that machine learning predictions should not be treated as perfect decisions. A student's actual academic outcome can also depend on factors that may not be completely represented in the dataset.

15. Results

The project results can be presented using tables and visualizations.

Important figures for the final project include:

1. Target class distribution
2. Exploratory data analysis visualizations
3. Model accuracy comparison
4. Model Macro F1 comparison
5. Final model evaluation results
6. Actual versus predicted outcomes, if generated

The model comparison clearly shows that SVM achieved the strongest overall results.

16. GitHub and Project Documentation

The project was organized using GitHub to maintain the project files and documentation.

The project documentation should contain:

- Project description

- Dataset information
- Data preparation process
- Machine learning workflow
- Models used
- Evaluation metrics
- Results
- Figures
- Requirements
- Final conclusions

Keeping the project organized makes it easier to understand, reproduce, and review.

17. Conclusion

This project applied machine learning classification techniques to predict student academic outcomes as Graduate, Dropout, or Enrolled.

The dataset contained 4,424 student records and 37 columns. The data audit found no missing values or duplicate rows. After preparing and exploring the data, three classification approaches were evaluated: a baseline model, KNN, and SVM.

The baseline achieved 49.94% accuracy and 22.21% Macro F1. KNN improved the results to 66.78% accuracy and 57.30% Macro F1. SVM achieved the best performance with 75.82% accuracy and 67.85% Macro F1.

Based on these results, SVM was the best-performing model among the tested models.

However, the model should be viewed as a decision-support tool rather than a final decision maker. Student outcomes can be influenced by many personal, academic, and social factors. Therefore, predictions should be interpreted carefully and should not replace human judgment or direct support from educational institutions.

Overall, the project demonstrates how machine learning can be used to analyze student data and provide useful predictions about academic outcomes.

18. Project Workflow Summary

Dataset → Data Audit → EDA → Train-Test Split → Preprocessing → Baseline → KNN → SVM → Model Comparison → Final Evaluation → Error Analysis → Results → Documentation